Anderson localisation with self-avoiding-walks

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1 Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 to explain the quantum mechanical effects of disorder in materials. Anderson discusses the behaviour of electrons in dirty crystals, which is the quantum mechanical analougue of a random walk in a random environment. The model considers a d-dimensional grid, in which each site represents an atom. From an heuristical point of view, we also consider electrons that jump from atom to atom and are subject to an external random potential, which influences their motion in the material. When these particles cannot jump , it indicates the absence of quantum transport.This phenomenon is called Anderson localisation. This model deals with conditions for which the electrons stay trapped in a specific region and don't generate electric conduction, defining an insulator. This is in sharp contrast to the behaviour in ideal crystals, which are always conductors.
Anderson's work, and that of N.F.Mott and J.H. van Vleck, won the 1977 physics Nobel prize for "their fundamental theoretical investigations of the electronic structure of magnetic and disordered systems". Since then, the study of random operators has become an important field of mathematical physics, which has led to a tremendous amount of research activity and many mathematical results.
The two important methods to prove Anderson localization are the *multiscale analysis* (MSA), developed by Froehlich and Spencer [] in 1983, and the *fractional moment method* (FMM) by Aizenman and Molchanov in 1993. The method of multiscale analysis, by now, has lead to a multitude of results on spectral and dynamical localization for a wide range of models. In this thesis we will focus on the fractional moment method, which has provided a simple perspective on localization and has enabled exponentially decaying bounds on expectation values of various propagation kernels.

structure of the thesis

Prerequisites

It is useful to have some familiarity with measure theory, basic probabilistic concepts such as indipendence and continuous distributions, and foundations of the theory of linear operators in Hilbert spaces, up to the spectral theorem for self adjoint operators and consequences (as spectral types).

1.1 The Anderson Model

Transport phenomena in a disordered material are often described via discrete random Schrödinger operator. In quantum mechanics, it corresponds to the classical total energy decomposing into kinetic and potential energy. On the lattice $\mathbb{Z}^d$, it takes the form of an infinite random matrix called the *Hamiltonian*:

$$H_\omega = -\Delta + \lambda V_\omega.$$

where $-\Delta$ is a deterministic matrix and λV_ω is a diagonal matrix where V_ω has random entries and λ is the disorder parameter.

This operator acts on the Hilbert space

$$\ell^2(\mathbb{Z}^d) = \{u : \mathbb{Z}^d \to \mathbb{C} : \sum_{x \in \mathbb{Z}^d} |u(x)|^2 < \infty\}$$

with inner product $\langle u, v\rangle = \sum_n \overline{u(x)} v(x)$.

In quantum mechanics, a physical state describing a single particle (such as an electron) is represented by a normalized wave function $\psi \in \ell^2(\mathbb{Z}^d)$ satisfying

$$\|\psi\|^2_{\ell^2} = \sum_{x \in \mathbb{Z}^d} |\psi(x)|^2 = 1. \tag{1.1}$$

The quantity $P(x) = |\psi(x)|^2$ defines a probability mass function on the lattice $\mathbb{Z}^d$: it represents the probability of finding the particle at site $x \in \mathbb{Z}^d$. This functions are the solutions to the time-evolving Schrödinger operator $H_\omega \phi(t) = i\partial_t \phi(t)$. Because the time evolution operator e^{-itH_ω} is unitary on $\ell^2(\mathbb{Z}^d)$, the total probability is strictly conserved at all times $t \in \mathbb{R}$:

$$\|\psi(t)\|^2_{\ell^2} = \|e^{-itH_\omega}\psi_0\|^2_{\ell^2} = \|\psi_0\|^2_{\ell^2} = 1. \tag{1.2}$$

In particular, if the particle is initially placed at a specific site $y \in \mathbb{Z}^d$ with $\psi(0) = \delta_y$, the transition probability to find the particle at site $x \in \mathbb{Z}^d$ at time t is given by the squared transition amplitude:

$$P(y \to x; t) = |\langle \delta_x, e^{-itH_\omega} \delta_y \rangle|^2. \tag{1.3}$$

As we'll see later on, under localisation conditions, P tends to 0 faster than any polinomial at all t for $|x - y| \to \infty$.

1.1.1 The Hamiltonian

The Descrete Laplacian is the descrete analogue of the usual negative Laplacian $(-\sum_j \partial^2/\partial x_j^2)$. For the functions $u \in \ell^2(\mathbb{Z}^d)$ it is defined as:

$$-\Delta u(x) := \sum_{k\in\mathbb{Z}^d, |x-k|=1} u(x) - u(k) \quad \text{for } x \in \mathbb{Z}^d. \tag{1.4}$$

where $|*|$ denotes the ℓ^2 norm.

The discrete Laplacian represents the kinetic term and connects neighbour sites on the lattice, such that the electron can transition from one to another.

For each site x, there are $2d$ possible k such that $|x-k| = 1$. Thus, we can define an infinte matrix such that:

$$(-\Delta)_{x,k} = \begin{cases} 2d & \text{if } x = k \quad \text{(on the diagonal)} \\ -1 & \text{if } |x-k| = 1 \\ 0 & \text{otherwise} \end{cases}$$

For $u \in \ell^2(\mathbb{Z}^d)$, $-\Delta u$ is summable in ℓ^2, bounded, and self-adjoint.

The first claim (summability) is shown by:

$$\begin{aligned} \sum_{x\in\mathbb{Z}^d} |\Delta u(x)|^2 &= \sum_{x\in\mathbb{Z}^d} \left| \sum_{k\in\mathbb{Z}^d, |x-k|=1} \big(u(x) - u(k)\big) \right|^2 \\ &\le 2d \sum_{x\in\mathbb{Z}^d} \left(\sum_{|x-k|=1} |u(x) - u(k)|^2 \right) \\ &\le 2d \sum_{x\in\mathbb{Z}^d} \left(2 \sum_{|x-k|=1} \big(|u(x)|^2 + |u(k)|^2\big) \right) \\ &\le 2d \cdot 2d \cdot 4 \|u\|_2^2 < \infty \end{aligned}$$

where we used the bound over the number of neighbors, triangle inequality, Cauchy Schwarz and the invariance under translation of the ℓ^2 norm. The boundedness comes from:

$$\frac{\|\Delta u\|_2^2}{\|u\|_2^2} \le 16d^2 \implies \|\Delta\|_2 \le 4d.$$

To prove self-adjointness, we only need to show symmetry (given $-\Delta$ is bounded). For $u, v \in \ell^2(\mathbb{Z}^d)$ and $k \in \mathbb{Z}^d$:

$$\langle u, -\Delta v\rangle = \sum_{x\in\mathbb{Z}^d} \overline{u(x)}(-\Delta v(x)) = \sum_{x\in\mathbb{Z}^d} \overline{u(x)} \left(\sum_{|x-k|=1} v(x) - v(k) \right)$$
$$= \sum_{x\in\mathbb{Z}^d} \left(\overline{u(x)} 2d v(x) - \sum_{|x-k|=1} \overline{u(x)} v(k) \right) = \sum_{x\in\mathbb{Z}^d} 2d\overline{u(x)} v(x) - \sum_{x\in\mathbb{Z}^d} \sum_{|x-k|=1} \overline{u(k)} v(x)$$
$$= \sum_{x\in\mathbb{Z}^d} v(x) \left(\sum_{|x-k|=1} \overline{u(x)} - \overline{u(k)} \right) = \sum_{x\in\mathbb{Z}^d} v(x) \overline{(-\Delta u(x))} = \langle -\Delta u, v\rangle.$$

The spectrum of a bounded, self-adjoint operator is real-valued, ad in this case equal to the closed interva $[0, 4d]$. In this range, we find the levels of kinetic energy that an electron owns. In order to understand this result better, let's consider the discrete Fourier transform:

$$\mathcal{F} \colon \ell^2(\mathbb{Z}^d) \longrightarrow L^2([0, 2\pi)^d)$$

such that:

$$(\mathcal{F}u)(x) \coloneqq \lim_{N\to\infty} (2\pi)^{-\frac{d}{2}} \sum_{|n|\le N} u(n)\, e^{-ix\cdot n}, \qquad x \in [0, 2\pi)^d.$$

We consider now calculations for $u \in \ell^1(\mathbb{Z}^d)$. For ℓ^1 is a dense subset of ℓ^2 and $\mathcal{F}$ unitary, we can extend the results by continuity for all $u \in \ell^2(\mathbb{Z}^d)$

$$
\begin{aligned}
(\mathcal{F}(-\Delta u))(x) &= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} (-\Delta u)(n)\, e^{-ix\cdot n} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \left(\sum_{|k-n|=1} \big(u(n)-u(k)\big) \right) e^{-ix\cdot n} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \left(2d\, u(n) - \sum_{|k-n|=1} u(k) \right) e^{-ix\cdot n} \\
&= 2d\,(\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \sum_{|k-n|=1} u(k)\, e^{-ix\cdot n} \\
&= 2d\,(\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{k\in\mathbb{Z}^d} u(k) \sum_{|m|=1} e^{-ix\cdot(k+m)} \quad (\text{for } m = n-k) \\
&= 2d\,(\mathcal{F}u)(x) - \left((2\pi)^{-\frac{d}{2}} \sum_{k\in\mathbb{Z}^d} u(k)\, e^{-ix\cdot k} \right) \sum_{|m|=1} e^{-ix\cdot m} \\
&= 2d\,(\mathcal{F}u)(x) - (\mathcal{F}u)(x) \sum_{j=1}^{d} \big(e^{-ix_j} + e^{ix_j}\big) \\
&= \left(2\sum_{j=1}^{d} (1-\cos(x_j)) \right) (\mathcal{F}u)(x).
\end{aligned}
$$

We obtain the multiplication operator:

$$
\mathcal{F}\Delta\mathcal{F}^{-1} = 2\sum_{j=1}^{d}(1-\cos(x_j))
$$

on $L^2([0,2\pi)^d)$ in the variable $x = (x_1, \ldots.., x_d)$. We notice that the Laplacian doesn't have eigenvalues in $\ell^2(\mathbb{Z}^d)$. Another similarity of Δ with the continuum Laplacian is that it has plane waves as generalized eigenfunctions. To see this, let $x \in [0,2\pi)^d$ and set

$$
\phi_x(n) := e^{-in\cdot x}
$$

While $\phi_x \notin \ell^2(\mathbb{Z}^d)$, Δ acts on it as

$$
\begin{aligned}
(\Delta\phi_x)(n) &= \sum_{|k|=1}(e^{-in\cdot x} - e^{-i(n+k)\cdot x}) \\
&= 2d\phi_x(n) - \left(\sum_{j=1}^{d} 2\cos(x_j)\right)\phi_x(n) \quad = \left(2\sum_{j=1}^{d}(1-\cos(x_j))\right)\phi_x.
\end{aligned}
$$

Thus ϕ_x is a bounded generalized normalized eigenfunction of Δ to the spectral value $2\sum_{j=1}^{d}(1-\cos(x_j))$. The spectrum is continuous and bounded in $[0, 2\dot{(}2d)]$.

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The Random Potential is expressed as the multiplication operator $V_\omega(x) = \omega_x$. It is a diagonal matrix, where $\omega = (\omega_x)_{x\in\mathbb{Z}^d}$ is the set of the indipendent identically distributed real-valued random entries. Physically, V_ω represents the random electric potential created by nuclei at the sites $n \in \mathbb{Z}^d$.

Definition 1.1 (i.i.d. random variables)**.** $(\omega_n)_{n\in\mathbb{Z}^d}$ are **identically distributed** if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_n \in A) = \mu(A)$ for all $n \in \mathbb{Z}^d$ and Borel sets $A \subset \mathbb{R}$.
$(\omega_n)_{n\in\mathbb{Z}^d}$ are **independent** if

$$
\mathbb{P}(\omega_{n_1} \in A_1, \ldots, \omega_{n_l} \in A_l) = \prod_{k=1}^{l} \mathbb{P}(\omega_{n_k} \in A_k) = \prod_{k=1}^{l} \mu(A_k)
$$

for each finite subset $\{n_1, \ldots, n_l\}$ of $\mathbb{Z}^d$ and arbitrary Borel sets $A_1, \ldots, A_l \subset \mathbb{R}$.

In this thesis we assume that ω_n are uniformly distributed in the real closed interval $[-1, 1]$. This means that $\mu(\omega_n) = \rho(\omega_n)d\omega_n$ where the continuous density is given by $\rho = \frac{1}{2}\chi_{[-1,1]}$.
It is possible to generalise the results to all absolutely continuous distributions where ρ has compact bounded support and to the Gaussian distribution.

As a diagonal operator, the spectrum of V_ω consists of the random entries ω_n, each corresponding to an eigenvector given by the Dirac delta $\delta_n \in \ell^2(\mathbb{Z}^d)$.

As for the descrete laplacian, the random term is summable in $\ell^2(\mathbb{Z}^d)$, bounded and self-adjoint. Self-adjointness of the maximal multiplication operator by a real-valued function is a fact. The spectrum is the closed interval $[-1, 1] \subset \mathbb{R}$, which means that V_ω is bounded, and since $\|V_\omega u\|^2_{\ell^2} \leq \|u\|^2_{\ell^2}$ it is also summable in $\ell^2(\mathbb{Z}^d)$.

The Disorder Parameter is represented by $\lambda > 0$. The value of λ is a great indicator for localisation, which makes it one of the central objects of interest of this thesis. Let's consider the two extreme cases:

- For $\boldsymbol{\lambda = 0}$ the Hamiltonian looses the random term and only considers the Laplacian. Without disorder, the material behaves like a perfect crystal without localisation. The state functions are continuous plane waves and the spectrum is also continuous between $[0, 4d]$.
- For $\boldsymbol{\lambda \gg 1}$ the kinetic hopping term becomes negligible compared to the random potential, so that $H_\omega \approx \lambda V_\omega$. In this scenario electrons are unable to hop between adjacent sites, and the material behaves as an insulator. In this limit, the system exhibits trivial localization: the eigenfunctions are localized Dirac deltas δ_x, and the spectrum is confined to the compact interval $[-\lambda, \lambda]$

The Green's Function definition

1.1.2 The spectrum

The next goal is to determine the spectrum of H_ω Because the random potential entries $(\omega_x)_{x\in\mathbb{Z}^d}$ are independent and identically distributed, the random Hamiltonian $H_\omega = -\Delta + \lambda V_\omega$ is an ergodic family of bounded self-adjoint operators. A standard consequence of ergodic operator theory is that the spectrum of H_ω is almost surely deterministic (see, for example, [**Pastur1980**; **Stolz2011**]): there exists a closed, non-random set $\Sigma \subset \mathbb{R}$ such that

$$\sigma(H_\omega) = \Sigma \qquad \text{for } \mathbb{P}\text{-almost all } \omega \in \Omega. \tag{1.5}$$

More specifically, as proven in [Theorem 2 GUNTERSTOLZ], the almost sure spectrum can be determined explicitly as the Minkowski sum of the spectrum of the kinetic energy operator and the support of the random potential distribution:

$$\Sigma = \sigma(-\Delta) + \lambda\,\mathrm{supp}(\mu). \tag{1.6}$$

Recall from Section **??** that under our non-negative convention for the discrete Laplacian, its spectrum is the band $\sigma(-\Delta) = [0, 4d]$. Since the single-site random variables ω_x are uniformly distributed on $[-1, 1]$, we have $\mathrm{supp}(\mu) = [-1, 1]$. Consequently, the almost sure spectrum of H_ω is given by the compact interval

$$\sigma(H_\omega) = [0, 4d] + [-\lambda, \lambda] = [-\lambda, 4d + \lambda] \qquad \text{almost surely.} \tag{1.7}$$

This confirms that the spectrum is a bounded subset of $\mathbb{R}$.

To characterize the physical nature of this spectrum and determine whether wave packets propagate or remain localized, we consider the spectral decomposition of the underlying Hilbert space. By the spectral theorem for self-adjoint operators (a detailed review of which is deferred to Appendix **??**), the Hilbert space $\ell^2(\mathbb{Z}^d)$ decomposes uniquely into an orthogonal direct sum of invariant subspaces:

$$\ell^2(\mathbb{Z}^d) = \mathcal{H}_{\mathrm{pp}} \oplus \mathcal{H}_{\mathrm{sc}} \oplus \mathcal{H}_{\mathrm{ac}}, \tag{1.8}$$

where:

- $\mathcal{H}_{\mathrm{pp}}$ is the subspace spanned by all eigenvectors of H_ω, corresponding to the *pure point spectrum* $\sigma_{\mathrm{pp}}(H_\omega)$;
- $\mathcal{H}_{\mathrm{ac}}$ is the subspace of states whose associated spectral measures are absolutely continuous with respect to Lebesgue measure, corresponding to the *absolutely continuous spectrum* $\sigma_{\mathrm{ac}}(H_\omega)$;
- $\mathcal{H}_{\mathrm{sc}}$ is the subspace corresponding to the *singular continuous spectrum* $\sigma_{\mathrm{sc}}(H_\omega)$.

In the case of self-adjoint bounded operators $A : \mathcal{H} \to \mathcal{H}$, the restriction to these subspaces defines the components of the spectrum:

$$\sigma(H_\omega) = \sigma_{\mathrm{pp}}(H_\omega) \cup \sigma_{\mathrm{ac}}(H_\omega) \cup \sigma_{\mathrm{sc}}(H_\omega). \tag{1.9}$$

From a physical perspective, states in $\mathcal{H}_{\mathrm{pp}}$ correspond to bound states that remain confined to finite spatial regions for all times, whereas states in the continuous subspace $\mathcal{H}_{\mathrm{c}} := \mathcal{H}_{\mathrm{ac}} \oplus \mathcal{H}_{\mathrm{sc}}$ represent scattering states associated with transport and wave-packet spreading. This heuristic correspondence is made mathematically rigorous by the RAGE theorem (named after Ruelle, Amrein, Georgescu, and Enss):

Theorem 1.2 (RAGE Theorem [**Cycon1987**; **HunzikerSigal2000**]). *Let H be a self-adjoint Schrödinger operator on $\ell^2(\mathbb{Z}^d)$. Then:*

(i) $\phi \in \mathcal{H}_{\mathrm{pp}}$ *if and only if*

$$\lim_{R\to\infty} \sup_{t\in\mathbb{R}} \|\chi_{\{|x|>R\}} e^{-itH}\phi\| = 0. \tag{1.10}$$

(ii) $\psi \in \mathcal{H}_{\mathrm{c}} = \mathcal{H}_{\mathrm{ac}} \oplus \mathcal{H}_{\mathrm{sc}}$ *if and only if for all $R > 0$,*

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \|\chi_{\{|x|\leq R\}} e^{-itH}\psi\|^2 \, dt = 0. \tag{1.11}$$

Here, $\chi_{\{|x|\leq R\}}$ and $\chi_{\{|x|>R\}}$ denote the projection operators corresponding to the characteristic functions of the ball $\{x \in \mathbb{Z}^d : |x| \leq R\}$ and its complement, respectively.

Theorem **??** confirms that for any bound state $\phi \in \mathcal{H}_{\mathrm{pp}}$, the time-evolved state $e^{-itH}\phi$ remains inside a sufficiently large ball uniformly in time up to an arbitrarily small error. Conversely, any scattering state in $\mathcal{H}_{\mathrm{c}}$ escapes every compact region in the time-mean. In terms of this classification, Anderson localization corresponds to the complete suppression of continuous spectrum, ensuring that $\sigma(H_\omega) = \sigma_{\mathrm{pp}}(H_\omega)$ almost surely.

1.2 Main results and Open Problems

As already stated before, there are two known approaches to the mathematical analysis of localization properties for random Schrödinger operators.

1.3 main results

In this section one could write first of all the main theorem and the spectral results. starting maybe with:
The aim of this thesis is to prove localisation on the whole spectrum of H_ω …
Poi scrivo il teorema finale con spiegazione che la dimostrazione avverrà nel capitolo 5.

potrebbe essere un'idea anche scrivere una piccola parentesi sulla differenza dei rislutati con Frohlich e spencer e con Aizenman e Molchanov.
One of the main results concerns the values of λ case for d=1,2,3

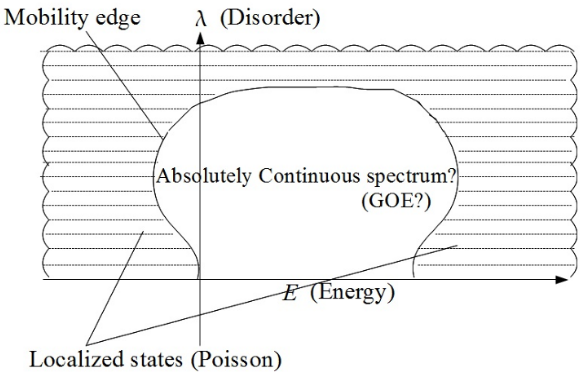

Figure 1: Aizenman figure

2 Localisation properties

We understood what localisation means from an heuristical point of view. Mathematically, it can be described from spectral and dynamical properties.

2.1 spectral localisation

Definition 2.1 (Spectral Localization)**.** random Schrödinger operator $H = H_\omega$ exhibits **spectral localization** if H_ω almost surely has only pure point spectrum, that is,

$$\sigma(H_\omega) = \sigma_{pp}(H_\omega) \quad \text{and} \quad \sigma_c(H_\omega) = \emptyset \quad \text{almost surely.}$$

Furthermore the eigenfunctions corresponding to all eigenvalues decay exponentially: for almost all ω, H_ω has a complete set of eigenfunctions $(\phi_{\omega,n})_{n\in\mathbb{N}}$ satisfying

$$|\phi_{\omega,n}(x)| \leq C_{\omega,n} e^{-\mu|x-x_{n,\omega}|}$$

where $\mu > 0$ is the localization length parameter, $x_{n,\omega}$ are the centers of localization, and $C_{\omega,n} < \infty$ is a finite constant.

2.1.1 Absence of transport

$$\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t) = \left\| \, |x| \, e^{-itH_\omega} P_{H_\omega \in [a,b]}\phi_0 \right\|^2 \tag{5}$$

If the electrons with energies in $[a, b]$ move ballistically with average velocity v_{av}, then roughly $x(t) \sim v_{av}t$ for large times. Hence $\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)$ will be proportional to t^2 for large t.

On the other hand, if the electrons in the energy interval $[a, b]$ are localized, then it should be natural to expect that $\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)$ is bounded uniformly in t. It is known (Simon, 1990) that pure point spectrum implies absence of ballistic motion,

$$\lim_{t\to\infty} \frac{\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)}{t^2} = 0$$

for all compactly supported initial conditions ϕ_0, as soon as the spectrum in $[a, b]$ is pure point. However, del Rio et al. (1996) constructed examples of (non-random) one-dimensional Schrödinger operators with pure point spectrum for all energies and exponentially localized eigenfunctions for which

$$\limsup_{t\to\pm\infty} \frac{\langle x^2\rangle_{\phi_0}(t)}{t^{2-\delta}} = \infty \qquad \text{for all } \delta > 0, \tag{6}$$

for a large class of compactly supported initial conditions ϕ_0. *Il fallimento risiede nella libertà delle costanti $C_{\omega,n}$: se $C_{\omega,n}$ cresce incontrollatamente con n, le code delle autofunzioni possono sovrapporsi su scale spaziali arbitrariamente grandi.*

2.2 dynamical localisation

Definition 2.2 (Dynamical localisation).

$$\sup_{t\in\mathbb{R}} \||X|^p e^{-itH_\omega}\psi\| < \infty \quad \textit{almost surely,} \tag{2.1}$$

where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$ and for all φ with compact support..

Definition 2.3 (Exponential localisation).

$$\mathbb{E}\left(\sup_{t\in\mathbb{R}} |\langle e_j, e^{-itH_\omega} e_k\rangle|\right) \leq Ce^{-\mu|j-k|} \tag{2.2}$$

Remark 2.4. Dynamical localization in the form (**??**) is a strong form of asserting that solutions of the time-dependent Schrödinger equation $H_\omega\phi(t) = i\partial\phi(t)$ are staying localized in space, uniformly for all times, and thus shows the absence of quantum transport.

Proposition 2.5. *Exponential localization implies that all moments of the position operator are bounded in time for almoste every realisation of V_ω, i.e., dynamical localisation.*

Proof. To see how (**??**) follows from (**??**), we assume that $\psi(k) = 0$ for $|k| > R$. Then

$$\begin{aligned}
\||X|^p e^{-itH_\omega}\psi\|^2 &= \sum_j |\langle e_j, |X|^p e^{-itH_\omega}\psi\rangle|^2 \\
&= \sum_j |j|^{2p} \left|\sum_{|k|\leq R} \langle e_j, e^{-itH_\omega} e_k\rangle \psi(k)\right|^2 \\
&\leq \sum_j \sum_{|k|\leq R} |j|^{2p} |\langle e_j, e^{-itH_\omega} e_k\rangle|^2 \|\psi\|^2,
\end{aligned}$$

where the last step used the Cauchy-Schwarz inequality. We can drop the square from $|\langle e_j, e^{-itH_\omega} e_k\rangle|^2$ (as this number is bounded by 1) and then take expectations to get

$$\begin{aligned}\mathbb{E}\left(\sup_t \||X|^p e^{-itH_\omega}\psi\|^2\right) &\leq \sum_j \sum_{|k|\leq R} |j|^{2p}\mathbb{E}\left(\sup_t |\langle e_j, e^{-itH_\omega} e_k\rangle|\right) \|\psi\|^2 \\ &\leq C \sum_j \sum_{|k|\leq R} |j|^{2p} e^{-\mu|j-k|} \|\psi\|^2 \\ &< \infty.\end{aligned}$$

□

Remark 2.6. The reason why we obtain dynamical localisation *almost surely*, is that we bound the expected value, instead of directly the norm.

Lemma 2.7. *Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space and let $A : \Omega \to [0, \infty]$ be a nonnegative, Σ-measurable random variable. Define $\Omega_0 := \{\omega \in \Omega \mid A(\omega) = \infty\}$. If $\mathbb{E}[A] < \infty$, then $\mathbb{P}(\Omega_0) = 0$; equivalently, $A < \infty$ $\mathbb{P}$-almost surely.*

Proof. For every $M > 0$, Markov's inequality gives $\mathbb{P}(A \geq M) \leq \frac{\mathbb{E}[A]}{M}$. Since $\Omega_0 \subseteq \{A \geq M\}$ for every M, we get $\mathbb{P}(\Omega_0) \leq \frac{\mathbb{E}[A]}{M} \to 0$ as $M \to \infty$, so $\mathbb{P}(\Omega_0) = 0$. □

2.3 From dynamical to spectral localisation

Proposition 2.8. *Suppose that dynamical localization in the form*

$$\mathbb{E}\left(\sup_{t\in\mathbb{R}} |\langle e_j, e^{-ith_\omega}\chi_I(h_\omega) e_k\rangle|\right) \leq Ce^{-\mu|j-k|} \tag{2.3}$$

holds in an open interval I. Then h_ω almost surely has pure point spectrum in I.

Proof. For a discrete Schrödinger operator $h = h_0 + V$ in $\ell^2(\mathbb{Z}^d)$ let $P_{\text{cont}}(h)$ be the projection onto its continuous spectral subspace. Then the RAGE-Theorem says that for every $\psi \in \ell^2(\mathbb{Z}^d)$,

$$\|P_{\text{cont}}(h)\chi_I(h)\psi\|^2 = \lim_{R\to\infty} \lim_{T\to\infty} \int_0^T \frac{dt}{T} \|\chi_{\{|x|\geq R\}} e^{-ith}\chi_I(h)\psi\|^2. \tag{2.4}$$

If ψ has finite support, say $\operatorname{supp}\psi \subset \{|x| \le r\}$, then

$$\begin{aligned}\|\chi_{\{|x|\ge R\}} e^{-ith}\chi_I(h)\psi\|^2 &\le \|\chi_{\{|x|\ge R\}} e^{-ith}\chi_I(h)\chi_{\{|x|\le r\}}\|^2\|\psi\|^2\\ &\le \|\chi_{\{|x|\ge R\}} e^{-ith}\chi_I(h)\chi_{\{|x|\le r\}}\|\|\psi\|^2\\ &\le \sum_{|x|\ge R, |y|\le r} |\langle e_x, e^{-ith}\chi_I(h)e_y\rangle|\|\psi\|^2,\end{aligned}$$

where dropping a square is allowed as $\|\chi_{\{|x|\ge R\}} e^{-ith}\chi_I(h)\chi_{\{|x|\le r\}}\| \le 1$.

Taking expectations in (2) implies, after using Fatou and Fubini,

$$\mathbb{E}\left(\|P_{\text{cont}}(h_\omega)\chi_I(h_\omega)\psi\|^2\right) \le \lim_{R\to\infty, T\to\infty} \int_0^T \frac{dt}{T} \sum_{|x|\ge R, |y|\le r} \mathbb{E}\left(|\langle e_x, e^{-ith_\omega}\chi_I(h_\omega)e_y\rangle|\right)\|\psi\|^2. \tag{2.5}$$

By (2.4) we have $\mathbb{E}\left(|\langle e_x, e^{-ith_\omega}\chi_I(h_\omega)e_y\rangle|\right) \le Ce^{-\mu|x-y|}$ uniformly in t, which bounds the right hand side of (2.6) by

$$\le \lim_{R\to\infty} \tilde{C} \sum_{|x|\ge R, |y|\le r} e^{-\mu|x-y|} = 0.$$

We conclude that $P_{\text{cont}}(h_\omega)\chi_I(h_\omega)\psi = 0$ for almost every ω and every ψ of finite support. The latter are dense in $\ell^2(\mathbb{Z}^d)$ and thus $P_{\text{cont}}(h_\omega)\chi_I(h_\omega) = 0$ almost surely, meaning that the spectrum in I is pure point. □

3 Fractional moments

In this section we are going to see the important lemmas and theorems that help us proving localisation. In particular we see the power of the fractional moment method.

3.1 The green function

The *resolvent operator* is defined as:

$$G_z = (H - z)^{-1}$$

3.2 Localisation at large disorder

Theorem 3.1 (Aizenman–Molchanov, 1993). *Let $0 < s < 1$. Then there exists $\lambda_0 > 0$ such that for $\lambda \geq \lambda_0$ there are $C < \infty$ and $\mu > 0$ with*

$$\mathbb{E}\left(|G_z(x,y)|^s\right) \leq Ce^{-\mu|x-y|} \tag{3.1}$$

uniformly in $x, y \in \mathbb{Z}^d$ and $z \in \mathbb{C} \setminus \mathbb{R}$. In particular, if λ_{And} satisfies $\lambda_{\text{And}} = \mu_d e \ln \lambda_{And}$ then for all $\lambda > \lambda_{\text{And}}, \lambda > e$ and $\epsilon > 0$ we have

$$\mathbb{E}\left(|G_z(x,y)|^{1-\frac{1}{\ln \lambda}}\right) \leq \beta(\lambda) K_\epsilon \ln \lambda \, e^{-m_\epsilon(\lambda)|x-y|} \tag{3.2}$$

where

$$m_\epsilon(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

Lemma 3.2 (A priori bound). *There exists a constant $C_1 = C_1(s,\rho) < \infty$ such that*

$$\mathbb{E}_x(|G_z(x,x)|^s) \leq C_1 \lambda^{-s} \tag{3.3}$$

for all $x \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$.

Here

$$C_1 = \frac{1}{1-s}$$

for $\omega_x \sim \mathcal{U}[-1,1]$

Lemma 3.3 (Resolvent identity). *For a linear operator $A : \mathcal{D}(A) \to \mathcal{H}$ and $z_1, z_2 \in \varrho(A)$,*

$$(A - z_1)^{-1} - (A - z_2)^{-1} = (A - z_1)^{-1}(z_1 - z_2)(A - z_2)^{-1}$$

Proof of a priori bound. This proof demonstrates the simplicity of the fundamental idea underlying the FMM. For fixed $x \in \mathbb{Z}^d$, write $\omega = (\hat{\omega}, \omega_x)$ where $\hat{\omega}$ is short for $(\omega_u)_{u \in \mathbb{Z}^d \setminus \{x\}}$. With $P_{e_x} := \langle e_x, \cdot \rangle e_x$, the orthogonal projection onto the span of e_x, we can separate the ω_x and $\hat{\omega}$ dependence of H_ω as

$$H_\omega = -\Delta + \lambda V_{\hat{\omega}} + \lambda \omega_x P_{e_x} = H_{\hat{\omega}} + \lambda \omega_x P_{e_x}.$$

For the operators $(H_\omega - z)$ and $(H_{\hat{\omega}} - z)$, the resolvent identity yields

$$(H_\omega - z)^{-1} = (H_{\hat{\omega}} - z)^{-1} - \lambda \omega_x (H_{\hat{\omega}} - z)^{-1} P_{e_x} (H_\omega - z)^{-1}.$$

(3.4)

Taking matrix-elements we conclude for the corresponding diagonal Green functions that

$$G_z(x, x) = \hat{G}_z(x, x) - \lambda \omega_x \hat{G}_z(x, x) G_z(x, x) \tag{3.5}$$

where $\hat{G}_z$ denotes the Green's function of $H_{\hat{\omega}}$. Collecting in $G_z(x, x)$:

$$G_z(x, x) = \frac{1}{a + \lambda \omega_x} \quad \text{with } a = \frac{1}{\hat{G}_z(x, x)}. \tag{3.6}$$

Note that the latter is well-defined since one can easily check that
$\mathrm{Im}\hat{G}_z(x, x) > 0$ for $\mathrm{Im} z > 0$.

The important fact is that a is a complex number which does not depend on ω_x. Thus, writing $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x)\, d\omega_x$ for $\rho(\omega_x) = \frac{1}{2}\chi_{[-1,1]}$, we find that

$$\mathbb{E}_x(|G_z(x, x)|^s) = \frac{1}{2\lambda^s} \int_{-1}^{1} \frac{d\omega_x}{|\frac{a}{\lambda} + \omega_x|^s} \leq \frac{1}{2\lambda^s} \int_{-1}^{1} \frac{d\omega_x}{|\omega_x|^s} = \frac{1}{(1-s)\lambda^s}, \tag{3.7}$$

with $\frac{1}{1-s}$ independent of λ and a, and thus independent of $\hat{\omega}$, z and x.

□

3.2.1 From fractional moment's exponential bound to dynamical localisation

In this subsection we aim to prove the following theorem:

Theorem 3.4. *Let $I \subset \mathbb{R}$ be an open bounded interval. If there exist $s \in (0,1)$, $C < \infty$ and $\mu > 0$ such that*

$$\mathbb{E}\left(|G_{E+i\eta}(x,y)|^s\right) \leq Ce^{-\mu|x-y|} \tag{3.8}$$

uniformly in $E \in I$ and $\eta > 0$, then dynamical localisation holds on the interval I.

Before diving in the proof, we need two helpers.

Lemma 3.5. *For a bounded and self-adjoint operator A,*

$$\left\|(A-z)^{-1}\right\| \leq \frac{1}{\mathrm{Im}(z)} \tag{3.9}$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof. For the definition of the operator norm, we know that

$$\left\|(A-z)^{-1}\right\| := \sup_{\mu \in \sigma((A-z)^{-1})} |\mu|$$

Let's consider a $\lambda \in \mathbb{C}$ and write $\mu = \frac{1}{\lambda - z}$. This means:

$$\begin{aligned}
\frac{1}{\lambda - z} \in \sigma((A-z)^{-1}) &\iff (A-z)^{-1} - (\lambda - z)^{-1} \text{ not invertible} \\
&\iff (A-z)^{-1}(A-\lambda)(\lambda-z)^{-1} \text{ not invertible} \\
&\iff (A-\lambda) \text{ not invertible} \\
&\iff \lambda \in \sigma(A)
\end{aligned}$$

Where the resolvent identity has been used in third Implication.
At this point we obtain

$$\left\|(A-z)^{-1}\right\| = \sup_{\lambda \in \sigma(A)} \frac{1}{|\lambda - z|} = \frac{1}{\inf_{\lambda \in \sigma(A)} |\lambda - z|} = \frac{1}{\mathrm{dist}(\sigma(A), z)}$$

For a bounded and self-adjoint operator, we know that the spectrum is real-valued. For $z \in \mathbb{C} \setminus \mathbb{R}$ the minimum distance between z and $\sigma(A)$ is the imaginary component of z:

$$\left\|(A-z)^{-1}\right\| \leq \frac{1}{|\mathrm{Im}(z)|} \tag{3.10}$$

□

Thid bound is one of the most important and useful in this thesis. Once again we applied the resolvent identity to find a simple yet useful result.

Proposition 3.6. *For every $s \in (0,1)$ there exists a constant $C_1 < \infty$ only depending on s and ρ such that*

$$|\operatorname{Im} z| \, \mathbb{E}_x \left(|G_z(x,y)|^2 \right) \leq C_1 \mathbb{E}_x \left(|G_z(x,y)|^s \right) \tag{3.11}$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$ and $x, y \in \mathbb{Z}^d$.

Proof. As in the proof of **??** we consider the Hamiltonian

$$H^{(\alpha)} = H_\omega + \alpha P_{e_x}.$$

Its Green function will be denoted by $G^{(\alpha)}$. Applying the resolvent identity, we find

$$(H_\omega - z)^{-1} = (H^{(\alpha)} - z)^{-1} + \alpha (H_\omega - z)^{-1} P_{e_x} (H^{(\alpha)} - z)^{-1}$$

and

$$\begin{aligned} G_z(x,y) &= G_z^{(\alpha)}(x,y) + \alpha G_z(x,y) G^{(\alpha)}(x,x) \\ \Rightarrow G_z^{(\alpha)}(x,y) &= \frac{G_z(x,y)}{1 + \alpha G_z(x,x)} \\ &= \frac{1}{\alpha + G_z(x,x)^{-1}} \cdot \frac{G_z(x,y)}{G_z(x,x)}. \end{aligned} \tag{3.12}$$

For the special case $x = y$ and $\tilde{\alpha} = -\operatorname{Re} G_z(x,x)^{-1}$ we get from (**??**) and Lemma **??** that

$$\left| \frac{1}{\operatorname{Im} G_z(x,x)^{-1}} \right| = |G_z^{(\tilde{\alpha})}(x,x)| \leq \frac{1}{|\operatorname{Im} z|},$$

i.e. $|\operatorname{Im} G_z(x,x)^{-1}| \geq |\operatorname{Im} z|$. Inserting this into (**??**) gives

$$|\operatorname{Im} z| \, |G_z^{(\alpha)}(x,y)|^2 \leq \frac{|\operatorname{Im} G_z(x,x)^{-1}|}{|\alpha + G_z(x,x)^{-1}|^2} \cdot \frac{|G_z(x,y)|^2}{|G_z(x,x)|^2}. \tag{3.13}$$

On the other hand, we can bound the same expression by

$$|\mathrm{Im}\, z|\, |G_z^{(\alpha)}(x,y)|^2 \leq |\mathrm{Im}\, z| \sum_{y' \in \mathbb{Z}^d} |G_z^{(\alpha)}(x,y')|^2$$
$$= |\mathrm{Im}\, z| \langle e_x\, (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1}\, e_x \rangle$$

For H_ω bounded and self adjoint, also $H^{(\alpha)}$ is bounded and self adjoint, da $\alpha \in \mathbb{R}$ and $\alpha < \infty$. It holds then,

$$|\mathrm{Im}\, z| \langle e_x\, (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1}\, e_x \rangle = |\mathrm{Im}\, z| \langle e_x,\, (H^{(\alpha)} - \overline{z})^{-1} (H^{(\alpha)} - z)^{-1}\, e_x \rangle$$

Applying the resolvent identity:

$$(H^{(\alpha)} - \overline{z})^{-1} - (H^{(\alpha)} - z)^{-1} = (H^{(\alpha)} - \overline{z})^{-1} (\mathrm{Im}\, (z)) (H^{(\alpha)} - z)^{-1}$$

. We obtain:

$$\begin{aligned}
&= |\mathrm{Im}\, z| \langle e_x, (H^{(\alpha)} - \overline{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle \\
&= |\mathrm{Im}\, z| \langle e_x, \frac{1}{\mathrm{Im}\, (z)} \left[(H^{(\alpha)} - z)^{-1} - (H^{(\alpha)} - \bar{z})^{-1} \right] e_x \rangle \\
&= |\mathrm{Im}\, G^{(\alpha)}(x,x;z)| \\
&= \frac{|\mathrm{Im}\, G_\omega(x,x;z)^{-1}|}{|\alpha + G_\omega(x,x;z)^{-1}|^2},
\end{aligned} \tag{3.14}$$

where the last step used (**??**) with $x = y$.

For $t \geq 0$ one has $\min(1, t^2) \leq t^s$, with $s \in (0,1)$. Using this to interpolate between (**??**) and (**??**) we get

$$|\mathrm{Im}\, z|\, |G^{(\alpha)}(x,y;z)|^2 \leq \frac{|\mathrm{Im}\, G_\omega(x,x;z)^{-1}|}{|\alpha + G_\omega(x,x;z)^{-1}|^2} \cdot \frac{|G_\omega(x,y;z)|^s}{|G_\omega(x,x;z)|^s}. \tag{3.15}$$

We will now use the following "re-sampling trick", which has the effect of creating an additional random variable (here α) to average over. For a nonnegative Borel function f on $\mathbb{R}$,

$$\begin{aligned}\iint f(\omega_x+\alpha)\rho(\omega_x+\alpha)\,d\alpha\,\rho(\omega_x)\,d\omega_x &= \iint f(\omega_x+\alpha)\rho(\omega_x+\alpha)\rho(\omega_x)\,d\omega_x\,d\alpha \\ &= \iint f(\omega_x)\rho(\omega_x)\rho(\omega_x-\alpha)\,d\omega_x\,d\alpha \\ &= \int f(\omega_x)\rho(\omega_x)\left(\int \rho(\omega_x-\alpha)\,d\alpha\right)d\omega_x \\ &= \int f(\omega_x)\rho(\omega_x)\,d\omega_x, \qquad (3.16)\end{aligned}$$

. where in the first and third steps we used Fubini, in the second we used translation invariance of Lebesgue measure and in the fourth we used the fact that $\int \rho(\omega_x-\alpha)\,d\alpha = 1$.

Choose $f(\omega_x) = |G_z(x,y)|^2$; then (**??**) and (**??**) yield

$$\begin{aligned}|\mathrm{Im}\, z|\,\mathbb{E}_x(|G_z(x,y)|^2) &= |\mathrm{Im}\, z|\,\mathbb{E}_x\left(\int |G_z^{(\alpha)}(x,y)|^2\rho(\omega_x+\alpha)\,d\alpha\right) \\ &\leq \mathbb{E}_x\left(|\mathrm{Im}\, G_z(x,x)^{-1}|\frac{|G_z(x,y)|^s}{|G_z(x,x)|^s}\int \frac{\rho(\omega_x+\alpha)}{|\alpha+G_z(x,x)^{-1}|^2}\,d\alpha\right). \qquad (3.17)\end{aligned}$$

We now use Lemma 3.7 below with $w = G_z(x,x)^{-1}$ to conclude

$$|\mathrm{Im}\, z|\,\mathbb{E}_x(|G_z(x,y)|^2) \leq C\mathbb{E}_x(|G_z(x,y)|^s),$$

with a constant $C<\infty$ which only depends on $\mathrm{supp}\,\rho$, but not on x, y and z. □

Lemma 3.7. *Let $0<s<1$. There exists a constant $C_s'<\infty$ such that*

$$|\mathrm{Im}\, w|\cdot|w|^s\int\frac{\rho(\omega_x+\alpha)}{|\alpha+w|^2}\,d\alpha \leq C_s'$$

uniformly in $w\in\mathbb{C}$ and $\omega_x \sim \mathcal{U}[-1,1]$.

Proof. The proof is that of [REFERENCE GUNTER STOLZ]. Since $|w|^s \leq |\alpha|^s + |\alpha+w|^s$, we bound:

1.

$$\begin{aligned}|\mathrm{Im}\, w|\int\frac{|\alpha|^s\rho(\omega_x+\alpha)}{|\alpha+w|^2}\,d\alpha &\leq |\mathrm{Im}\, w|\int\frac{\||\alpha|^s\rho(\omega_x+\alpha)\|_\infty}{|\alpha+w|^2}\,d\alpha \\ &\leq |\mathrm{Im}\, w|\,\||\alpha|^s\rho(\omega_x+\alpha)\|_\infty\int\frac{1}{(\alpha+\mathrm{Re}\, w)^2+\mathrm{Im}\, w^2}\,d\alpha\end{aligned}$$

for $a, b \in \mathbb{R}$ we have

$$\int \frac{1}{a^2+b^2}\, da = 2\pi i \operatorname{Res}\left(\frac{1}{a^2+b^2}, ib\right) = \frac{2\pi i}{2ib}.$$

Thus for $\rho = \frac{1}{2}\chi_{[-1,1]}$ and $|\alpha| \leq 1 + |\omega| \leq 2$

$$\begin{aligned} |\operatorname{Im} w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2}\, d\alpha &\leq \pi \| |\alpha|^s \rho(\omega_x + \alpha)\|_\infty \\ &\leq \pi 2^{s-1}. \end{aligned}$$

2. Now, we bound the second expression.

   (a) For $|\operatorname{Im} w|$ big:
   Since $|\alpha + w| \geq |\mathrm{Im} w|$ and $\int \rho(\omega_x + \alpha)\, d\alpha = 1$:

   $$I_2 \leq |\mathrm{Im} w| \cdot \frac{1}{|\mathrm{Im} w|^{2-s}} \int \rho(\omega_x + \alpha)\, d\alpha = \frac{1}{|\mathrm{Im} w|^{1-s}}$$

   (b) For $|\operatorname{Im} w|$ small:

   $$\begin{aligned} |\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}}\, d\alpha &\leq |\operatorname{Im} w| \frac{1}{2} \int \frac{d\alpha}{|\alpha + w|^{2-s}} \\ &= |\operatorname{Im} w| \frac{1}{2} \int \frac{1}{|u + \mathrm{Im} w|^{2-s}}\, du \\ &= |\operatorname{Im} w|^2 \frac{1}{2} \int \frac{1}{|u^2 + \mathrm{Im} w^2|^{\frac{2-s}{2}}}\, du \\ &= \frac{1}{2} (\operatorname{Im} w)^s \int \frac{1}{|t^2 + 1|^{\frac{2-s}{2}}}\, dt \end{aligned}$$

   by setting $u = \alpha + \operatorname{Re} w$ and $t = \frac{u}{\operatorname{Im} w}$.
   To establish the finiteness of the constant, we consider the integral

   $$C_s := \int_{-\infty}^{\infty} \frac{1}{(t^2+1)^{\frac{2-s}{2}}}\, dt.$$

   For $s \in (0,1)$, the exponent satisfies $2 - s > 1$, which ensures the asymptotic decay $(t^2+1)^{-\frac{2-s}{2}} \sim |t|^{-(2-s)}$ as $|t| \to \infty$, rendering the integral absolutely convergent.Then we have

$$|\mathrm{Im}\, w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}}\, d\alpha \leq \min\left(\frac{1}{|\mathrm{Im}\, w|^{1-s}}, C_s \frac{1}{2} |\mathrm{Im}\, w|^s \right) \leq C_s' 2^{s-1}.$$

□

Now we can procede with the proof of the theorem.

Proof of Theorem 3.4. For H_ω bounded and self-adjoint, from the spectral theorem, we define the complex Borel spectral measure $\mu_{\delta_x,\delta_y} = \mu_{x,y}$ such that:

$$\mu_{x,y}(B) := \langle \delta_x, \chi_B(H_\omega)\delta_y \rangle.$$

with B Borel in $\mathbb{R}$. The total variation measure $|\mu_{x,y}|$ over the open bounded interval I satisfies

$$\begin{aligned} |\mu_{x,y}|(I) &= \sup_{\substack{g:\mathbb{R}\to\mathbb{C} \text{ Borel} \\ |g|\leq 1}} \left| \int g(\lambda)\, d\mu_{x,y}(\lambda) \right| \\ &= \sup_{|g|\leq 1} |\langle \delta_x, g(H_\omega)\chi_I(H_\omega)\delta_y \rangle|, \end{aligned} \tag{3.18}$$

e.g. [Theorem 6.16 Rudin] for p=1. This implies:

$$\sup_{t\in\mathbb{R}} |\langle \delta_x, e^{-itH_\omega}\chi_I(H_\omega)\delta_y \rangle| \leq |\mu_{x,y}|(I).$$

From Lusin's theorem, we can extend the bound also for $g \in C_c(I)$:

$$|\mu_{x,y}|(I) = \sup_{\substack{g\in C_c(I) \\ |g|\leq 1}} |\langle \delta_x, g(H_\omega)\delta_y \rangle|.$$

da I bounded open interval. For $g \in C_c(I)$ it follows by elementary analysis (using that g is bounded and uniformly continuous) that, uniformly in $\lambda \in \mathbb{R}$,

$$g(\lambda) = \lim_{\eta\downarrow 0} \frac{\eta}{\pi} \int \frac{g(E)}{(\lambda - E)^2 + \eta^2}\, dE.$$

By the spectral theorem and dominated convergence, this implies:

$$\begin{aligned}
\langle e_x, g(H_\omega)e_y\rangle &= \int_{\mathbb{R}} g(\lambda)\, d\mu_{x,y}(\lambda) \\
&= \lim_{\eta\downarrow 0} \int_{\mathbb{R}} \left(\frac{\eta}{\pi} \int_I \frac{g(E)}{(\lambda - E)^2 + \eta^2}\, dE \right) d\mu_{x,y}(\lambda) \\
&= \lim_{\eta\downarrow 0} \frac{\eta}{\pi} \int_I g(E) \langle \delta_x, (H_\omega - E - i\eta)^{-1}(H_\omega - E + i\eta)^{-1}\delta_y\rangle\, dE
\end{aligned}$$

Now we can bound $\mathbb{E}(|\mu_{x,y}|)$, considering that $|g(E)| \leq 1$:

$$\begin{aligned}
\mathbb{E}(|\mu_{x,y}|(I)) &\leq \mathbb{E}\left(\lim_{\eta\downarrow 0} \frac{\eta}{\pi} \int_I g(E) \langle \delta_x, (H_\omega - E - i\eta)^{-1}(H_\omega - E + i\eta)^{-1}\delta_y\rangle\, dE \right) \\
&\leq \mathbb{E}\left(\lim_{\eta\to 0^+} \frac{\eta}{\pi} \int_I \sum_{z\in\mathbb{Z}^d} |\langle \delta_x, (H_\omega - E - i\eta)^{-1}\delta_z\rangle| |\langle \delta_z, (H_\omega - E + i\eta)^{-1}\delta_y\rangle|\, dE \right) \\
&\leq \lim_{\eta\to 0^+} \frac{1}{\pi} \int_I \sum_{z\in\mathbb{Z}^d} \left(\mathbb{E}(\eta |\langle \delta_x, (H_\omega - E - i\eta)^{-1}\delta_z\rangle|^2) \right)^{1/2} \\
&\quad \cdot \left(\mathbb{E}(\eta |\langle \delta_z, (H_\omega - E + i\eta)^{-1}\delta_y\rangle|^2) \right)^{1/2} dE,
\end{aligned}$$

where, in this order, Fatou, Fubini and Cauchy–Schwarz (on $\mathbb{E}$) have been used. Now we can use Proposition **??** and the hypothesis to obtain:

$$\begin{aligned}
\mathbb{E}(|\mu_{x,y}|(I)) &\leq \frac{C_1}{\pi} \int_I \sum_{z\in\mathbb{Z}^d} \left(\mathbb{E}|G_\omega(x,z;E+i\eta)|^s\right)^{1/2} \left(\mathbb{E}|G_\omega(z,y;E-i\eta)|^s\right)^{1/2} dE \\
&\leq \frac{C_1|I|C}{\pi} \sum_{z\in\mathbb{Z}^d} e^{-\frac{\mu}{2}|x-z|} e^{-\frac{\mu}{2}|z-y|}.
\end{aligned}$$

By the triangle inequality, $\frac{\mu}{2}|x-z| + \frac{\mu}{2}|z-y| \geq \frac{\mu}{4}|x-y| + \frac{\mu}{4}|x-z| + \frac{\mu}{4}|z-y|$. Therefore,

$$\mathbb{E}(|\mu_{x,y}|(I)) \leq \left(\frac{C_1|I|C}{\pi} \sum_{z\in\mathbb{Z}^d} e^{-\frac{\mu}{4}|x-z|} e^{-\frac{\mu}{4}|z-y|} \right) e^{-\frac{\mu}{4}|x-y|} \leq \widetilde{C} e^{-\widetilde{\mu}|x-y|},$$

with $\widetilde{\mu} = \frac{\mu}{4}$ and finite prefactor $\widetilde{C} = \frac{C_1|I|C}{\pi} \sum_{u\in\mathbb{Z}^d} e^{-\frac{\mu}{4}|u|}$. □

From (**??**) we can also easily obtain spectral localisation through Simon-Wolff criterion [book or paper].

Proposition 3.8 (Simon-Wolff criterion). *Let H_ω be a random operator in*

$l^2(\mathbb{Z}^d)$ with a random potential whose conditional-single site distribution is absolutely continuous. Then if for Lebesgue-almost every $E \in \mathbb{R}$ and $\mathbb{P}$-almost all ω:

$$\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x,y)|^2 < \infty \tag{3.19}$$

H_ω has only pure point spectrum.

Proof can be found in [BOOK]

Theorem 3.9. *Let Theorem **??** hold. Then, the spectrum of H_ω consists only of pure point spectrum for a.e. realisation of V_ω.*

The proof, recalls the one proposed in [DISERTORI]

proof of Theorem 4.9. From (**??**) we notice that for all $\lambda > \lambda_{\text{And}}$:

$$\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x,y)|^s] = C \sum_{y \in \mathbb{Z}^d} e^{-\mu|x-y|} < \infty \quad \text{for all } \mu > 0 \tag{3.20}$$

uniformly for $z \in \mathbb{C} \setminus \mathbb{R}$ and $x \in \mathbb{Z}^d$ fixed.

From Preposition **??** we notice also that $\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x,y)|^2]$ always exists but might be infinite for $\eta \to 0$. We argue:

$$\begin{aligned}
\mathbb{E}\left[\left(\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x,y)|^2\right)^{\frac{s}{2}}\right] &= \mathbb{E}\left[\lim_{\eta \downarrow 0}\left(\sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x,y)|^2\right)^{\frac{s}{2}}\right] \\
&\leq \mathbb{E}\left[\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x,y)|^s\right] \\
&\leq \liminf_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x,y)|^s] < \infty,
\end{aligned}$$

where in the first two steps we used that the function $x \mapsto x^{\frac{s}{2}}$ is monotone and $(\sum_n a_n)^s \leq \sum_n a_n^s$ for all $a_n \geq 0$ and $0 < s < 1$. The last two inequalities follow by Fatou and (**??**). From Lemma **??** we notice that

$$\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x,y)|^2 < \infty \qquad \text{almost surely}$$

. □

4 SAW representation

4.1 SAW Definitions

Definition 4.1 (Self-Avoiding Walk)**.** A **self-avoiding walk** (SAW) on the integer lattice $\mathbb{Z}^d$ of length $N \in \mathbb{N}_0$ is an ordered $(N+1)$-tuple of distinct lattice sites

$$\gamma = (\gamma_0, \gamma_1, \ldots, \gamma_N) \in (\mathbb{Z}^d)^{N+1}$$

such that consecutive vertices are nearest neighbors, that is,

$$|\gamma_i - \gamma_{i-1}| = 1 \quad \text{for all } i \in \{1, \ldots, N\},$$

and $\gamma_i \neq \gamma_j$ for all $i \neq j$.

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Definition 4.2 (Sets of Self-Avoiding Walks)**.** Let $\Lambda \subseteq \mathbb{Z}^d$. For any $x, y \in \Lambda$ and $N \in \mathbb{N}_0$:

(i) $\mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N contained entirely in Λ, starting at $\gamma_0 = x$ and terminating at $\gamma_N = y$. When $\Lambda = \mathbb{Z}^d$, we simply write $\mathcal{S}_N(y, x)$.

(ii) $\mathcal{S}_N^{(\Lambda)}(x) := \bigcup_{y \in \Lambda} \mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N in Λ starting at x.

(iii) $c_N := |\mathcal{S}_N(x)| = \sum_{y \in \mathbb{Z}^d} |\mathcal{S}_N(y, x)|$ denotes the total number of self-avoiding walks of length N on $\mathbb{Z}^d$ originating from a fixed site (e.g., the origin 0), which is independent of the starting vertex by translation invariance.

Definition 4.3 (Connectivity Constant)**.** The **connective constant** (or connectivity constant) μ_d of the lattice $\mathbb{Z}^d$ is defined as the asymptotic growth rate of c_N:

$$\mu_d := \lim_{N \to \infty} (c_N)^{1/N}.$$

By subadditivity of $\log c_N$, this limit exists, is positive, and satisfies the strict inequality $\mu_d < 2d - 1$.

Claim 4.4. Let μ_d be the connective constant (or growth rate) of self-avoiding walks on $\mathbb{Z}^d$. Then:

$$\mu_d \leq 2d - 1$$

Proof. Let c_N denote the number of self-avoiding walks (SAWs) of length N starting from the origin $0 \in \mathbb{Z}^d$, and let $S_N(0,x)$ denote the set of such walks ending at $x \in \mathbb{Z}^d$. By definition:

$$c_N = \sum_{x\in\mathbb{Z}^d} |S_N(0,x)| = \sum_{x\in\mathbb{Z}^d} \sum_{\gamma\in S_N(0,x)} 1$$

Consider the generating function (*susceptibility*) with parameter $\xi \in \mathbb{C}$:

$$\sum_{N=0}^{\infty} |\xi|^N c_N = \sum_{N=0}^{\infty} |\xi|^N \sum_{x\in\mathbb{Z}^d} |S_N(0,x)|$$

For $N = 0$, $c_0 = 0$. For $N \geq 1$, a walk cannot immediately step back onto the previous vertex (non-reversing condition). Thus:

- The first step has at most $2d$ choices.
- Each of the subsequent $N-1$ steps has at most $2d-1$ choices.

This gives the upper bound:

$$c_N \leq 2d(2d-1)^{N-1}, \quad \text{for } N \geq 1.$$

Using this upper bound, we can bound the series from above:

$$\begin{aligned}
\sum_{N=0}^{\infty} |\xi|^N c_N &\leq \sum_{N=1}^{\infty} |\xi|^N (2d)(2d-1)^{N-1} \\
&= 2d|\xi| \sum_{k=0}^{\infty} |\xi|^k (2d-1)^k \quad \text{(setting } k = N-1\text{)} \\
&= 2d|\xi| \sum_{k=0}^{\infty} (|\xi|(2d-1))^k
\end{aligned}$$

This geometric series converges if and only if:

$$|\xi|(2d-1) < 1 \iff |\xi| < \frac{1}{2d-1} =: R$$

where $R = \frac{1}{2d-1}$ is the radius of convergence of the bounding series.

By the Cauchy–Hadamard theorem, the radius of convergence R' of the original series $\sum_{N=0}^{\infty} c_N |\xi|^N$ is:

$$R' = \frac{1}{\limsup_{N\to\infty} c_N^{1/N}} = \frac{1}{\mu_d}$$

Since the original series is bounded term-by-term by a series that converges for $|\xi| < R$, the true radius of convergence must satisfy:

$$R' \geq R \implies \frac{1}{\mu_d} \geq \frac{1}{2d-1}$$

Taking the reciprocal yields:

$$\mu_d \leq 2d - 1.$$

□

Definition 4.5 (SAW Correlation Function)**.** For a parameter $\beta \geq 0$, the **self-avoiding walk correlation function** $C_\beta(y-x)$ between two vertices $x, y \in \mathbb{Z}^d$ is the generating function

$$C_\beta(y-x) := \sum_{N=0}^{\infty} \beta^N \, |\mathcal{S}_N(y,x)|,$$

defined for all values of β for which the power series is absolutely convergent. By translation invariance, $C_\beta(y-x)$ depends only on the displacement vector $y-x$.

Definition 4.6 (SAW Susceptibility)**.** The **susceptibility** $\chi(\beta)$ of self-avoiding walks is the sum of the two-point correlation function over all end-points:

$$\chi(\beta) := \sum_{x \in \mathbb{Z}^d} C_\beta(x) = \sum_{N=0}^{\infty} c_N \, \beta^N.$$

The series converges uniformly absolutely if and only if $0 \leq |\beta| 1/\mu_d$, where $1/\mu_d$ is the critical radius of convergence.

Proposition 4.7 (Exponential Decay of the Correlation Function)**.** *Let $\beta \geq 0$ be such that $\beta < 1/\mu_d$. Then for every $\varepsilon > 0$ there exists a constant $K_\varepsilon < \infty$, independent of x, such that*

$$C_\beta(x) \;\leq\; K_\varepsilon \big((\mu_d + \varepsilon)\beta\big)^{|x|} \qquad \textit{for all } x \in \mathbb{Z}^d.$$

Proof. We know that $|\mathcal{S}_N(0,x)| = 0$ whenever $N < |x|$, and the sum defining $C_\beta(x)$ may be restricted to $N \geq |x|$:

$$C_\beta(x) = \sum_{N=|x|}^{\infty} \beta^N \, |\mathcal{S}_N(x,0)| \;\leq\; \sum_{N=|x|}^{\infty} \beta^N c_N. \tag{4.1}$$

Fix $\varepsilon > 0$. Since $\mu_d = \lim_{N\to\infty} c_N^{1/N}$, by the definition of limit there exists $N_0 \in \mathbb{N}$ such that

$$c_N^{1/N} < \mu_d + \varepsilon \qquad \text{for all } N \geq N_0,$$

that is,

$$c_N < (\mu_d + \varepsilon)^N \qquad \text{for all } N \geq N_0.$$

For $N = 0, 1, \ldots, N_0 - 1$, define

$$K_\varepsilon \ := \ \max\left(1, \ \max_{0 \leq N < N_0} \frac{c_N}{(\mu_d + \varepsilon)^N}\right),$$

which is finite, being the maximum of finitely many finite quantities. By construction it holds:

$$c_N \ \leq \ K_\varepsilon (\mu_d + \varepsilon)^N \qquad \text{for all } N \geq 0.$$

Combining it with (4.1) we obtain

$$C_\beta(x) \ \leq \ K_\varepsilon \sum_{N=|x|}^{\infty} \left[(\mu_d + \varepsilon)\beta\right]^N.$$

Choosing ε small enough that $r := (\mu_d + \varepsilon)\beta < 1$ (possible since $\beta < 1/\mu_d$), the tail of the geometric series equals

$$\sum_{N=|x|}^{\infty} r^N = \frac{r^{|x|}}{1-r}.$$

Hence

$$C_\beta(x) \ \leq \ \frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta} \left[(\mu_d + \varepsilon)\beta\right]^{|x|}.$$

where $\dfrac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta}$ is still finite and x-independent. $\square$

4.2 SAW representation of G

The proof of the theorem 3.1 is similar to the one given by Schenker in []. We are going to work with the Green's function defined on a volume where one site is removed every stage. Therefore , for any $\Lambda \supset \mathbb{Z}^d$ (finite or infinite) the Green function changes:

$$G^{\Lambda}(x,y;z) := \begin{cases} (H^{\Lambda}_{\omega} - z), & \forall x,y \in \Lambda \\ 0 & \text{, otherwise} \end{cases} \tag{4.2}$$

where $H^{\Lambda}_{\omega} - z \in \mathbb{C}^{\Lambda\times\Lambda}$is the matrix obtained by restricting $H_{\omega} - z$ to Λ. In this thesis we will consider $\Lambda = \mathbb{Z}^d$, but we leave the noation Λ to highligth the fact that the same result holds for any volume.
To notice is also that $H^{\Lambda}_{\omega} - z$ is still invertible for all $z \in \mathbb{C} \setminus \mathbb{R}$. In this section, for semplicity during calculations, we will write $G^{\Lambda}(x,y)$ instead of $G^{\Lambda}(x,y;z)$.

Theorem 4.8. *Let $x,y \in \mathbb{Z}^d$, $x \neq y$, $\Lambda \subseteq \mathbb{Z}^d$. G^{Λ} is defined as in (4.1). It holds*

$$G^{\Lambda}(x,y) = \sum_{n=1}^{N} \sum_{\gamma\in\mathcal{S}_n(x,y)} \prod_{i=1}^{n} G^{\Lambda}(x,x)\cdot G^{\Lambda_i}(\gamma_i,\gamma_i) + R_N \tag{4.3}$$

where $\Lambda_i = \Lambda \setminus \{x,\gamma_1,\dots,\gamma_{i-1}\}$ *and*

$$R_N = \sum_{x\sim y} \sum_{\gamma\in\mathcal{S}_N(0,x)} \prod_{i=1}^{N-1} G^{\Lambda}(x,x)\cdot G^{\Lambda_i}(\gamma_i,\gamma_i)\cdot G^{\Lambda_N}(\gamma_N,y). \tag{4.4}$$

Claim 4.9.

$$G^{\Lambda}(x,y) = G^{\Lambda}(x,x)\sum_{x_1\sim x} G^{\Lambda_1}(x_1,y) \tag{4.5}$$

Proof. Using the resolvent identity for $A = (H^{\Lambda}_{\omega} - z)$ and $B = (H^{\{x\}}_{\omega} \oplus H^{\Lambda\setminus\{x\}}_{\omega} - z)$, we obtain:

$$\begin{aligned}&(H^{\Lambda}_{\omega} - z)^{-1} - (H^{\{x\}}_{\omega} \oplus H^{\Lambda\setminus\{x\}}_{\omega} - z)^{-1} \\ &= -(H^{\Lambda}_{\omega} - z)^{-1}\left(H^{\Lambda}_{\omega} - (H^{\{x\}}_{\omega} \oplus H^{\Lambda\setminus\{x\}}_{\omega})\right)(H^{\{x\}}_{\omega} \oplus H^{\Lambda\setminus\{x\}}_{\omega} - z)^{-1}\end{aligned}$$

We know:

$$\begin{aligned} H^{\Lambda}_{\omega} &= 2d - \sum_{x_1\sim y}\delta_{x_1} + \lambda V_{\omega} \quad \forall y \in \Lambda \\ H^{\{x\}}_{\omega} &= \lambda\omega_x + 2d \\ H^{\Lambda\setminus\{x\}}_{\omega} &= 2d - \sum_{x_1\sim y}\delta_{x_1} + \lambda V_{\omega} - \lambda\omega_x \quad \forall y \in \Lambda\setminus\{x\} \end{aligned}$$

$$\implies H^\Lambda_\omega - (H^{\{x\}}_\omega \oplus H^{\Lambda\setminus\{x\}}_\omega) = -\sum_{x_1\sim x} \delta_{x_1} =: T_{x,x_1}$$

Now considering the matrix elements in (x,y) with $x \neq y$ and matrix multiplication:

$$(H^\Lambda_\omega - z)^{-1}_{x,y} - 0 = -\sum_{x_1,x_2} (H^\Lambda_\omega - z)^{-1}_{x,x_1} T_{x_1,x_2} (H^{\{x\}}_\omega \oplus H^{\Lambda\setminus\{x\}}_\omega - z)^{-1}_{x_2,y}$$

For $T_{x_1,x_2} \neq 0$, we need $x_1 = x$ and $x_2 \sim x_1$.

We obtain:

$$G^\Lambda(x,y) = \sum_{x_1\sim x} G^\Lambda(x,x) G^{\Lambda\setminus\{x\}}(x_1,y)$$

□

Proof of Theorem 5.8. Now we separate the case in which $x_1 = y$ (which implies $|x-y| = 1$).

$$G^\Lambda(x,y) = G^\Lambda(x,x)\left(\sum_{x_1\sim x, x_1\neq y} G^{\Lambda_1}(x_1,y) + I_{|x-y|=1}\, G^{\Lambda_1}(y,y)\right)$$

Iterating N times, it gives (**??**). □

4.3 Proof with SAW

Now we use self-avoiding-walks to prove the theorem (3.1).

Proof. We take the average of (4.2), and using $|\sum a|^s \leq \sum |a|^s$ (which follows from triangle inequality), we have:

$$\mathbb{E}\left[|G^\Lambda(x,y)|^s\right] \leq \sum_{n=1}^{N} \sum_{\gamma\in\mathcal{S}_n(x,y)} \mathbb{E}\left[|G^\Lambda(x,x)|^s \prod_{i=1}^{n} |G^{\Lambda_i}(\gamma_i,\gamma_i)|^s\right] + \tag{4.6}$$

$$\sum_{x\sim y} \sum_{\gamma\in\mathcal{S}_N(0,x)} \mathbb{E}[|G^\Lambda(x,x)|^s \prod_{i=1}^{N-1} |G^{\Lambda_i}(\gamma_i,\gamma_i)|^s \cdot |G^{\Lambda_N}(\gamma_N,y)|^s.] \tag{4.7}$$

The resolvent $G^{\Lambda_j}(w,w')$ does not depend on the random variables ω_{x_i}, $i = 0,\ldots,j-1$. Hence, recursive applications of the a priori bound (3.2) yield

$$\mathbb{E}\left[|G^{\Lambda}(x,x)|^s\prod_{j=1}^{n}|G^{\Lambda_j}(\gamma_j,\gamma_j)|^s\right]\leq\left(\frac{1}{1-s}\frac{1}{\lambda^s}\right)^{n+1}$$

and

$$\mathbb{E}\left[|G^{\Lambda}(x,x)|^s\prod_{j=1}^{N-1}|G^{\Lambda_j}(\gamma_j,\gamma_j)|^s\cdot|G^{\Lambda_{N-1}}(\gamma_N,y)|^s\right]$$
$$\leq\left(\frac{1}{1-s}\cdot\frac{1}{\lambda^s}\right)^{N}\cdot\mathbb{E}\left[|G^{\Lambda_{N-1}}(\gamma_N,y)|^s\right]\leq\left(\frac{1}{1-s}\frac{1}{\lambda^s}\right)^{N}\frac{1}{|\mathrm{Im}(z)|^s}$$

Where we used a priori bound and lemma 3.5.

Inserting these estimates in the sums above, we get:

$$\mathbb{E}\left(|G^{(\Lambda)}(x,y)|^s\right)\leq\sum_{n=0}^{N}\beta(s)^{1+n}|\mathcal{S}_n^{(\Lambda)}(y,x)|+Err(N)\frac{1}{|\mathrm{Im}z|^s}\tag{4.8}$$

with $Err(N):=\beta(s)^N|\mathcal{S}_N^{(\Lambda)}(x)|$ $\beta(s)=\frac{1}{1-s}\frac{1}{\lambda^s}$ Let's consider the susceptability $\chi(\beta)$ of the self avoiding walks in $\mathcal{S}_N^{(\Lambda)}(x)$:

$$\chi(\beta)=\sum_{x\in\mathbb{Z}^d}\sum_{N=0}^{\infty}\beta(s)^N\,|S_N^{(\Lambda)}(x)|=\sum_{N=0}^{\infty}c_N\,\beta(s)^N.\tag{4.9}$$

The series converges absolutely if and only if $0\leq|\beta|<1/\mu_d$, where $1/\mu_d=\lim_{N\to\infty}c_N^{1/N}$ is the critical radius of convergence. Thus, for $|\beta(s)|=\frac{1}{|(1-s)\lambda^s|}\leq\frac{1}{\mu_d}$, $\chi(\beta)$ converges uniformly, and most importantly:

$$Err(N)\longrightarrow 0,\quad\text{for}\quad N\to\infty$$

We send N to the infinity and obtain from (4.7):

$$\mathbb{E}\left(|G^{(\Lambda)}(x,y)|^s\right)\leq\sum_{n=0}^{\infty}\beta(s)^{1+n}|\mathcal{S}_n^{(\Lambda)}(y,x)|\tag{4.10}$$

Taking the supremum over volumes Λ now yields

$$\sup_{\Lambda\subset\mathbb{Z}^d}\mathbb{E}\left(|G^{(\Lambda)}(x,y)|^s\right)\leq\beta(s)C_{\beta(s)}(x-y).\tag{4.3}$$

It remains to establish for which λ we may find $s \in (0,1)$ with $\beta(s) < 1/\mu_d$. As above, for $\lambda > e$ the unique critical point is $s_{\text{crit}} = 1 - 1/\ln\lambda$ and

$$\min_{s\in(0,1)} \beta(s) = \beta(s_{\text{crit}}) = \frac{e\ln\lambda}{\lambda}$$

which is less than $1/\mu_d$ if and only if $\lambda > \lambda_{\text{And}}$. Finally we get:

$$\mathbb{E}\left(|G^{(\Lambda)}(x,y;z)|^{1-\frac{1}{\ln\lambda}}\right) \leq \beta(\lambda) K_\epsilon \ln\lambda\, e^{-m_\epsilon(\lambda)|x-y|} \tag{4.11}$$

where

$$m_\epsilon(\lambda) = -\ln\beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

□

5 Connection to Long-Range SAW

The purpose of this chapter is to extend the self-avoiding walk (SAW) representation of the resolvent, to random Hamiltonians with long-range hopping. In particular, we compare this setting with the previous model and apply the fractional moment method to study localization. The difference lies in the kinetic operator, which is replaced by the fractional descrete Laplacian $(-\Delta)^{\alpha}$ with $0 < \alpha < 1$. This change leads to a polynomial decay bound for the avareged Green's function, in sharp contrast with the standard exponential decay. This chapter is based on the work of M. Disertori, R.M. Escobar and C. Rojas-Molina in 2024 [].

5.1 The fractional Laplacian

For the long-range, the kinetic operator also connects arbitrarily distant sites in the lattice, rather than just the first neighbors.

Definition 5.1. Let $-\Delta$ be defined as in 1.4 in Section 1. The fractional descrete Laplacian is $(-\Delta)^{\alpha}$ for $0 < \alpha < 1$

Just as the standard descrete Laplacian, $(-\Delta)^{\alpha}$ is bounded, self-adjoint and translation invariant. Furthermore, it satisfies

$$(-\Delta)^{\alpha}_{x,x} > 0 \quad \forall x \in \mathbb{Z}^d, \qquad (-\Delta)^{\alpha}_{x,y} \leq 0 \quad \forall x \neq y \text{ in } \mathbb{Z}^d$$

Moreover, its off-diagonal matrix elements exhibit polynomial decay: there exist constants $0 < c_{\alpha,d} \leq \mathcal{C}_{\alpha,d} < \infty$ such that

$$\frac{c_{\alpha,d}}{|x-y|^{d+2\alpha}} \leq -(-\Delta)^{\alpha}_{x,y} \leq \frac{C_{\alpha,d}}{|x-y|^{d+2\alpha}} \tag{5.1}$$

From this bound, one deduces the summability of fractional powers of the matrix elements:

Lemma 5.2. *For any $\alpha \in (0,1)$ and $s \in \left(\frac{d}{d+2\alpha}, 1\right]$,*

$$\sum_{x \in \mathbb{Z}^d} |(-\Delta)^{\alpha}_{0,x}|^s < \infty \tag{5.2}$$

Proof. From (**??**):

$$\sum_{|x|\geq 1,\, x\in\mathbb{Z}^d} |(-\Delta)^\alpha_{0,x}|^s \leq C^s_{\alpha,d} \sum_{|x|\geq 1,\, x\in\mathbb{Z}^d} \frac{1}{|x|^{(d+2\alpha)s}}.$$

By standard comparison of the lattice sum with an integral on $\mathbb{R}^d$:

$$\sum_{|x|\geq 1,\, x\in\mathbb{Z}^d} \frac{1}{|x|^p} \leq C_d \int_{|z|>1} \frac{1}{|z|^p}\, dz.$$

Passing to spherical coordinates in $\mathbb{R}^d$, the integral reads

$$\begin{aligned}\int_{|z|>1} \frac{1}{|z|^p}\, dz &= |\mathbb{S}^{d-1}| \int_1^\infty \frac{r^{d-1}}{r^p}\, dr \\ &= |\mathbb{S}^{d-1}| \int_1^\infty r^{d-1-p}\, dr,\end{aligned}$$

where $|\mathbb{S}^{d-1}| = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ denotes the surface area of the unit sphere in $\mathbb{R}^d$.

The radial integral converges if and only if the power satisfies

$$d-1-p < -1 \iff p > d.$$

Substituting $p = s(d+2\alpha)$, this condition becomes

$$s(d+2\alpha) > d \iff s > \frac{d}{d+2\alpha}.$$

Since $\alpha > 0$ implies $d + 2\alpha > d$; for every $s \in \left(\frac{d}{d+2\alpha}, 1\right]$, we conclude that

$$\sum_{x\neq 0} |(-\Delta)^\alpha_{0,x}|^s \leq C^s_{\alpha,d} \sum_{x\neq 0} \frac{1}{|x|^{s(d+2\alpha)}} < \infty.$$

$\square$

5.2 The SAW Representation for Long-Range Resolvents

We consider the fractional Anderson Hamiltonian

$$H_{\alpha,\omega} = (-\Delta)^\alpha + \lambda V_\omega \quad \text{on } \ell^2(\mathbb{Z}^d), \tag{5.3}$$

with the same definitions given in Chapter 2. In this case we generalise directly for an arbitrary continuous distribution with compact supported

density.
The Green's function is again the resolvent and its decay is close to the decay of the fractional Laplacian.

As proven in [theorem 1 of DISERTORI], the averaged fractional Green's function satisfies

$$\mathbb{E}[|G_z(x,y)|^s] \leq \frac{K_0\theta_s}{\lambda^s}\frac{1}{|x-y|^{d+2\alpha_s}} \qquad \forall\, x,y \in \mathbb{Z}^d,\ x \neq y, \tag{5.4}$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$, for all $\lambda > \lambda_0(s) \coloneqq (\frac{\theta_s}{R})^{\frac{1}{s}}$, where $R_{\chi(\beta,\alpha)}$ is the radius of convergence of the susceptability and with $2\alpha_s \coloneqq s(d+2\alpha)-d$. In addition, $K_0 = K_0(d,\alpha)$ comes from the two-point function bound and $\frac{\theta_s}{\lambda^s}$ comes from the *a priori bound*.

Remark 5.3. Notice that for $\alpha = 1$ and considering the uniform distribution in $[-1,1]$ for ω, $\theta_s = \frac{1}{1-s}$ which leads to $\lambda_0(s) = \lambda_{\text{And}}$.

In the long-range scenario, we are still using the same proof techniques as in Chapter 5 in order to prove (**??**).
Nonetheless, one important change is given by the fact that, for long jumps, we are not just considering the neighbors, but an arbitrary site on the lattice. In particular, the probability to jump from the site x to the site y changes into:

$$p(x,y) \coloneqq \frac{|(-\Delta)^{\alpha}_{x,y}|^s}{\sum_{z\neq x}|(-\Delta)^{\alpha}_{x,z}|^s}.$$

Using the random walk on $\mathbb{Z}^d$ with $p(x,y)$ as transition probability, changes the definitions given in Chapter 5.1, which are now all generated by $|(-\Delta)^{\alpha}_{x,y}|^s$ and well defined for the values of s as in (**??**).

In order to prove (**??**) we use the same proof steps as before:

1. As in Claim **??**, by the resolvent identity we obtain:

$$G^{\Lambda}_z(x,y) = G^{\Lambda}_z(x,x) \sum_{x_1 \in \Lambda\setminus\{x\}} \Delta^{\alpha}_{x,x_1} G^{\Lambda_1}(x_1,y) \tag{5.5}$$

with $\Delta^{\alpha} = -(-\Delta)^{\alpha}$ greater than zero for the off-diagonal elements. Since we are working with $0 < \alpha < 1$, we don't obtain an operator such as T_{x,x_1} but we keep the Laplacian fractional.

2. We then iterate and bound like in Chapter 5 for (**??**), in order to get:

$$\mathbb{E}[|G_z^{\Lambda}(x,y)|^s] \leq \frac{\theta_s}{\lambda^s} C_{\beta}^{\alpha,s}(y-x) + \frac{1}{|\operatorname{Im} z|^s} Err(N). \tag{5.6}$$

   In this case, $Err(N)$ depends on $|(-\Delta)^{\alpha}_{x,y}|^s$ and under the conditions of (**??**) it vanishes for $N \to \infty$, solving the problem of the $\operatorname{Im} z$ as denominator in the equation.

3. Finally, (see [Lemma 4 of PAPER DISERTORI]),

$$C_{\beta}^{\alpha,s}(y-x) \leq K_0 \frac{1}{|y-x|^{d+2\alpha_s}} \tag{5.7}$$

   holds as a consequence of (**??**).

5.3 The Localisation critiria

A natural question that may arise from the previuous statements is how localisation is proven without exponental decay of the fractional moments of the Green's function. In the long-range case it is only possible to prove a weaker version of dynamical localisation. Without finitness for all moments of the position operator, it is convenient to prove spectral localisation via Simon-Wolff theorem.

Lemma 5.4. *Let's* (**??**) *hold uniformly in* $z \in \mathbb{C} \setminus \mathbb{R}$ *and for* $\lambda > \lambda_0(s)$. *Then it follows:*

(i) The spectrum of $H_{\alpha,\omega}$ *consists only of pure point spectrum for a.e.* $\omega \in \Omega$

(ii) for $0 < \beta < 2\alpha$ *we have for a.e.* $\omega \in \Omega$

$$\sup_{t\in\mathbb{R}} \| |X|^{\beta/2} e^{-itH_{\omega,\alpha}} \delta_0 \|^2 < \infty. \tag{5.8}$$

where X *denotes the position operator as in* (**??**)*.In particular, if* $\alpha > \frac{1}{2}$ *the first moment of the wave function initially localized at the origin and evolving under the action of* $H_{\alpha,\omega}$ *is finite.*

Proof. (i) From (**??**) we obtain:

$$\sum_{x\in\mathbb{Z}^d} \mathbb{E}[|G_z(0,x)|^s |x|^t] \leq \frac{K_0\,\theta_s}{\lambda^s} \sum_{x\in\mathbb{Z}^d} \frac{1}{|x|^{d+2\alpha_s - t}} < \infty \quad \forall t \in [0, 2\alpha_s) \tag{5.9}$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ and for all $\lambda > \lambda_0(s)$. We continue as in the proof of Theorem **??**. At the end we apply the Simon-Wolff criterion and obtain spectal localisation almost everywhere.

(ii) Proof in [PAPER DISERTORI]

$\square$

5.4 The SAW Representation for Long-Range Resolvents

We consider the fractional Anderson Hamiltonian

$$H_{\alpha,\omega} = (-\Delta)^{\alpha} + \lambda V_{\omega} \quad \text{on } \ell^2(\mathbb{Z}^d), \tag{5.10}$$

where $V_{\omega}(x) = \omega_x$ is an independent and identically distributed random field with single-site distribution P_0. We assume P_0 is τ-regular for some $\tau \in \left(\frac{d}{d+2\alpha}, 1\right]$ with regularity constant $M_{\tau}(P_0)$ (Definition 3.1).

For $s \in \left(\frac{d}{d+2\alpha}, \tau\right)$, we define the exponent

$$2\alpha_s := s(d+2\alpha) - d > 0. \tag{5.11}$$

Let $D(x,y) := |(-\Delta)^{\alpha}(x,y)|^s$ for $x \neq y$ serve as the step-weight kernel of a long-range self-avoiding walk. Let χ_D^{SAW} denote its susceptibility, with radius of convergence $R_{\chi_D^{\mathrm{SAW}}}$, and let $C_{D,\gamma}^{\mathrm{SAW}}(x)$ be the two-point correlation function with activity $\gamma > 0$.

Theorem 5.5 ([**Disertori2024**]). *Define*

$$\theta_s := \frac{\tau}{\tau - s} M_{\tau}(P_0)^{\frac{s}{\tau}}, \qquad \lambda_0(s) := \left(\frac{\theta_s}{R_{\chi_D^{\mathrm{SAW}}}}\right)^{\frac{1}{s}}. \tag{5.12}$$

For any $\lambda > \lambda_0(s)$, setting $\gamma = \theta_s \lambda^{-s} < R_{\chi_D^{\mathrm{SAW}}}$, the fractional moment of the Green's function satisfies

$$\mathbb{E}[|G_z(x,y)|^s] \leq \gamma\, C_{D,\gamma}^{\mathrm{SAW}}(x-y) \leq \frac{K_0 \theta_s}{\lambda^s} \frac{1}{|x-y|^{d+2\alpha_s}} \qquad \forall\, x \neq y, \tag{5.13}$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$, where $K_0 = K_0(d, \alpha, s, \lambda) < \infty$.

Outline of Proof. The proof proceeds via the depleted resolvent strategy adapted to non-local hoppings:

1. **Depleted Resolvent Expansion:** Let $\Lambda \subset \mathbb{Z}^d$ and write $\Delta^{\alpha}(x,y) := -(-\Delta)^{\alpha}(x,y) > 0$ for $x \neq y$. By the resolvent identity, isolating the jump to y gives

$$G_z^{\Lambda}(x,y) = G_z^{\Lambda}(x,x)\Delta^{\alpha}(x,y)G_z^{\Lambda\setminus\{x\}}(y,y) + G_z^{\Lambda}(x,x) \sum_{w_1 \in \Lambda\setminus\{x,y\}} \Delta^{\alpha}(x,w_1) G_z^{\Lambda\setminus\{x\}}(w_1,y). \tag{5.14}$$

Iterating this expansion N times generates self-avoiding paths $\{w_0, w_1, \ldots, w_n\}$ of lengths $n \leq N$.

2. **Averaging via the A-Priori Bound:** Because $G_z^{\Lambda\setminus\{w_0,\ldots,w_{j-1}\}}(w_j, w_j)$ is independent of the random variables at visited sites, conditional integration using the a-priori bound $\mathbb{E}_{w_j}[|G^\Lambda(w_j, w_j)|^s] \le \theta_s \lambda^{-s}$ factors across path vertices:

$$\mathbb{E}[|G_z^\Lambda(x,y)|^s] \le \frac{\theta_s}{\lambda^s} C_{D,\gamma}^{\mathrm{SAW}}(x-y) + \mathrm{Err}(N). \tag{5.15}$$

   Since $\lambda > \lambda_0(s)$, the activity satisfies $\gamma < R_{\chi_D^{\mathrm{SAW}}}$, ensuring that $\lim_{N\to\infty} \mathrm{Err}(N) = 0$.

3. **Decay of the SAW Two-Point Function:** As established by Chen and Sakai [**CS15**], the two-point function of a self-avoiding walk whose jump kernel decays as $|x|^{-(d+2\alpha_s)}$ satisfies $C_{D,\gamma}^{\mathrm{SAW}}(x) \le K_0|x|^{-(d+2\alpha_s)}$ for all $\gamma < R_{\chi_D^{\mathrm{SAW}}}$, yielding (**??**).

□

5.5 Spectral and Dynamical Consequences

Because $(-\Delta)^\alpha$ decays algebraically, the fractional moments of the Green's function decay polynomially rather than exponentially. Consequently, moments of the position operator $|X|^p$ are bounded only for limited powers p, precluding full dynamical localization. Spectral localization, however, remains accessible via the Simon–Wolff criterion.

Theorem 5.6 ([**Disertori2024**]). *Assume P_0 is τ-regular and $\lambda > \lambda_0(s)$ for some $s \in \left(\frac{d}{d+2\alpha}, \tau\right)$. Then:*

(i) ***Spectral Localization:*** *The spectrum of $H_{\alpha,\omega}$ is almost surely pure point:*

$$\sigma_{\mathrm{cont}}(H_{\alpha,\omega}) = \emptyset \qquad a.s. \tag{5.16}$$

(ii) ***Partial Dynamical Localization:*** *If P_0 has a bounded density ($\tau = 1$), then for every $\beta \in (0, 2\alpha)$,*

$$\sup_{t\in\mathbb{R}} \sum_{x\neq 0} |x|^\beta |e^{-itH_{\alpha,\omega}}(0,x)|^2 < \infty \qquad a.s. \tag{5.17}$$

In particular, when $\alpha > 1/2$, the first spatial moment ($\beta = 1$) is uniformly bounded in time.

Proof. (i) By Theorem **??**, for any $t \in [0, 2\alpha_s)$,

$$\sum_{x\in\mathbb{Z}^d} \mathbb{E}[|G_z(0,x)|^s |x|^t] \leq \frac{K_0\theta_s}{\lambda^s} \sum_{x\neq 0} \frac{1}{|x|^{d+2\alpha_s-t}} < \infty, \tag{5.18}$$

uniformly in $z \in \mathbb{C}\setminus\mathbb{R}$. Setting $t = 0$, monotonicity of $u \mapsto u^{s/2}$, the elementary subadditivity inequality $(\sum_n a_n)^{s/2} \leq \sum_n a_n^{s/2}$, and Fatou's lemma yield

$$\begin{aligned}\mathbb{E}\left[\left(\lim_{\eta\downarrow 0}\sum_{y\in\mathbb{Z}^d}|G_{E+i\eta}(x,y)|^2\right)^{\frac{s}{2}}\right] &\leq \mathbb{E}\left[\liminf_{\eta\downarrow 0}\sum_{y\in\mathbb{Z}^d}|G_{E+i\eta}(x,y)|^s\right]\\ &\leq \liminf_{\eta\downarrow 0}\sum_{y\in\mathbb{Z}^d}\mathbb{E}[|G_{E+i\eta}(x,y)|^s] < \infty.\end{aligned} \tag{5.19}$$

This implies $\sum_y |G_{E+i\eta}(x,y)|^2 < \infty$ for Lebesgue-almost every energy $E \in \mathbb{R}$ with probability one. By the Simon–Wolff theorem, the spectrum of $H_{\alpha,\omega}$ is purely pure point.

(ii) The bound follows from controlling the eigenfunction correlator $Q(x, 0; \Sigma)$ via the energy integral of the fractional Green's function, combined with the summability of $|x|^\beta |x|^{-(d+2\alpha_s)}$ for s chosen sufficiently close to 1 so that $\beta < 2\alpha_s < 2\alpha$.

□